

JAI VADULA

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Summary

Energetic and innovative AI Engineer eager to leverage a profound background in AI, specializing in LLM frameworks, and cloud deployments. Aims to apply technical skills and creativity in developing state-of-the-art AI solutions within a dynamic team setting.

Skills

Programming Languages: Python, C++

Frameworks & Libraries: Lang-Chain, Hugging-Face, TensorFlow, LLM, Pandas, NumPy, Matplotlib, Scikit-Learn

AI/ML Techniques: Fine-tuning with custom data, vector embedding, NLP, neural network optimization, MLOPS, Dockers

Developer Tools: Git, GitHub, Jupyter, VS Code

Soft Skills: Analytical thinking, problem-solving, teamwork, effective communication

Experience

AI Engineer Intern

October 2025 – Present

Danfoss Industries Pvt Ltd

Chennai, India (On-site)

- Developed a rule-based Machine Learning pipeline using Python to automatically generate Playwright test scripts from JSON inputs, covering both positive and negative test scenarios across 2 core modules — Alarm and Schedules.
- Built a hybrid Decision Tree Classifier (Scikit-learn) pipeline that auto-generated 188+ Playwright test cases across four alarm modules, reducing manual test authoring effort by approximately 70%.
- Designed and implemented scalable Azure DevOps analytics using WQL and REST APIs, building and operationalizing 8+ engineering metrics — including Test Automation Coverage, Test Coverage, Defect Acceptance Rate, Say-Do Ratio, Team Capacity, Pull Request Compliance, Unit Test Code Coverage, and Static Code Analysis — across Power BI dashboards adopted by 9 teams across Danfoss globally.
- Enabled Stakeholders, Senior Managers, and Senior Product Managers to validate and review monthly of 9 teams through a single, reliable Power BI reporting interface, bringing visibility and accountability to engineering quality standards across the organization.
- Engineered a transition from OData extraction to REST API batch processing, reducing Power BI refresh time from 3+ hours (often failing due to Power BI Service's 7,199 second timeout limit) down to under 2 minutes — a 98% reduction in data refresh time.

Projects

DBPal — Natural Language to SQL Interface

- Overview: Developed a conversational AI interface that enables non-technical users to query MySQL and PostgreSQL databases using natural language prompts, bridging the gap between raw data and business intelligence.
- Technologies: Python, LangChain, Streamlit, Scikit-learn, Statsmodels, Plotly, Altair, NumPy, Pandas.
- Outcome: Eliminated the need for SQL expertise among business users by automating query translation, and delivered real-time interactive dashboards with automated statistical analysis including time-series forecasting and trend detection.

MediVoxAI — Multimodal Healthcare AI Assistant

- Overview: Designed a multimodal AI healthcare assistant capable of simultaneously analyzing medical imagery and patient queries, enabling hands-free medical consultations through an end-to-end voice processing pipeline.
- Technologies: Python, Llama 3 Vision, Groq Cloud, OpenAI Whisper, ElevenLabs, gTTS, Gradio, Hugging Face Spaces.
- Outcome: Demonstrated a scalable proof-of-concept for real-time diagnostic support automation, achieving high-speed inference via Groq Cloud and seamless voice interaction through accurate transcription and realistic text-to-speech synthesis.

GuestGazer — Hotel Cancellation Prediction System

- Overview: Built an end-to-end MLOps pipeline for hotel cancellation prediction, enabling data-driven business strategies such as dynamic pricing, overbooking optimization, and proactive fraud detection.
- Technologies: Python, Scikit-learn, MLflow, Flask, Docker, Jenkins, Google Cloud Run, Google Cloud Storage, Google Container Registry, NumPy, Pandas, PyYAML, GitHub.
- Outcome: Automated the full model lifecycle from training to production deployment using CI/CD via Jenkins and Docker containerization, delivering a user-friendly web interface that transforms predictive analytics into actionable revenue management insights.

Certifications

- Oracle Certified Generative AI Professional - Oracle
- GitHub Foundations Certification – GitHub
- Machine Learning Specialization – Coursera

Education

Integrated Masters of Technology in Computer Science and Engineering

2021 – 2026

Specialization in Artificial Intelligence and Machine Learning

CGPA: 8.86 / 10

Vellore Institute of Technology (VIT)